

The Jahn-Teller Theorem

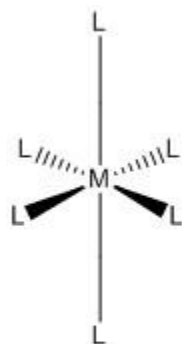
- The Jahn-Teller Theorem (named after Hermann Arthur Jahn and Edward Teller), was published in 1937 .

- means that:-
- any non-linear molecular system in a degenerate electronic state will be unstable and will undergo distortion to form a system of lower symmetry and lower energy thereby removing the degeneracy.

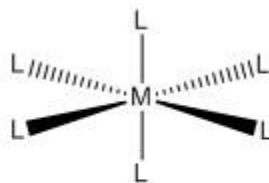
- The Jahn-Teller effect is a geometric distortion of a non-linear molecular system that reduces its symmetry and energy.
- This distortion is typically observed among octahedral complexes where the two axial bonds can be shorter or longer than those of the equatorial bonds.
- This effect can also be observed in tetrahedral compounds.
- This effect is dependent on the electronic state of the system.

- For a given octahedral complex, the five d atomic orbitals are split into two degenerate sets when constructing a molecular orbital diagram.
- These are represented by the sets' symmetry labels: t_{2g} (d_{xz} , d_{yz} , d_{xy}) and e_g (d_z^2 and $d_{x^2-y^2}$).
- When a molecule possesses a degenerate electronic ground state, it will distort (Jahn-Teller effect) to remove the degeneracy and form a lower energy (and by consequence, lower symmetry) system.
- The octahedral complex will either elongate or compress the z ligand bonds as shown below:

- In an octahedral crystal field, the t_{2g} orbitals occur at lower energy than the e_g orbitals.
- This is a reflection of the orientation of the orbitals since the t_{2g} are directed between bond axes while the e_g point along bond axes.
- The shielding effect this has on the electrons is used to explain why the Jahn-Teller effect is generally only important for odd number occupancy of the e_g level.
- The effect of Jahn-Teller distortions is best documented for Cu(II) complexes (with 3 electrons in the e_g level) where the result is that most complexes are found to have elongation along the z-axis.



Elongated



Compressed

- *Jahn-Teller distortions for an octahedral complex.*

- When an octahedral complex exhibits elongation, the axial bonds are longer than the equatorial bonds.
- For a compression, it is the reverse; the equatorial bonds are longer than the axial bonds.
- Elongation and compression effects are dictated by the amount of overlap between the metal and ligand orbitals.
- Thus, this distortion varies greatly depending on the type of metal and ligands.
- In general, the stronger the metal-ligand orbital interactions are, the greater the chance for a Jahn-Teller effect to be observed.

Elongation

- Elongation Jahn-Teller distortions occur when the degeneracy is broken by the stabilization (lowering in energy) of the d orbitals with a z component, while the orbitals without a z component are destabilized (higher in energy) as shown below

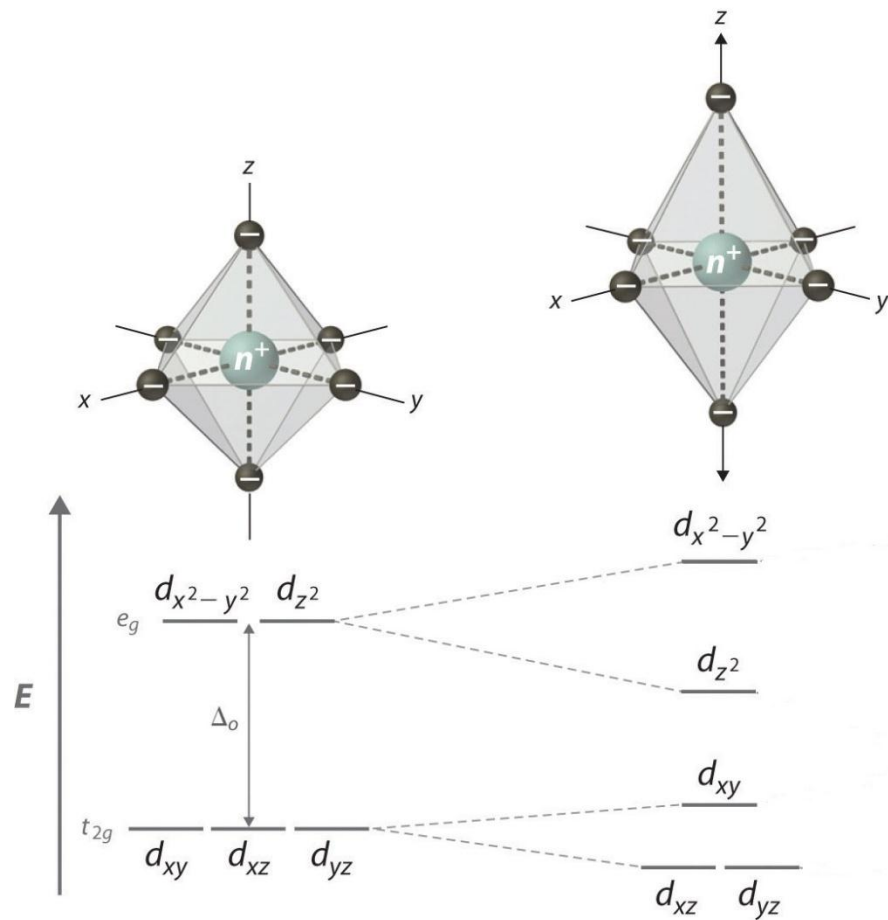
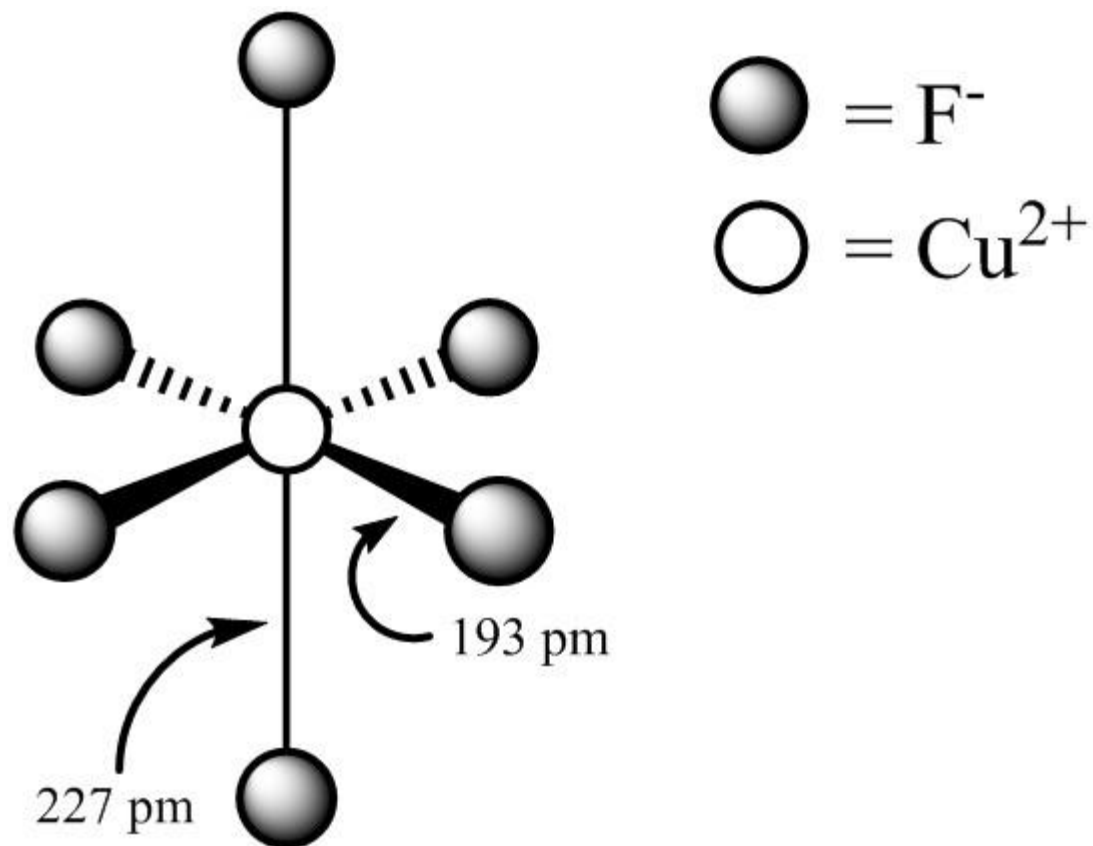


Illustration of tetragonal distortion (elongation) for an octahedral complex.

- This is due to the d_{xy} and $d_{x^2-y^2}$ orbitals having greater overlap with the ligand orbitals, resulting in the orbitals being higher in energy.
- Since the $d_{x^2-y^2}$ orbital is antibonding, it is expected to increase in energy due to elongation.
- The d_{xy} orbital is still nonbonding, but is destabilized due to the interactions.
- Jahn-Teller elongations are well-documented for copper(II) octahedral compounds. A classic example is that of copper(II) fluoride as.

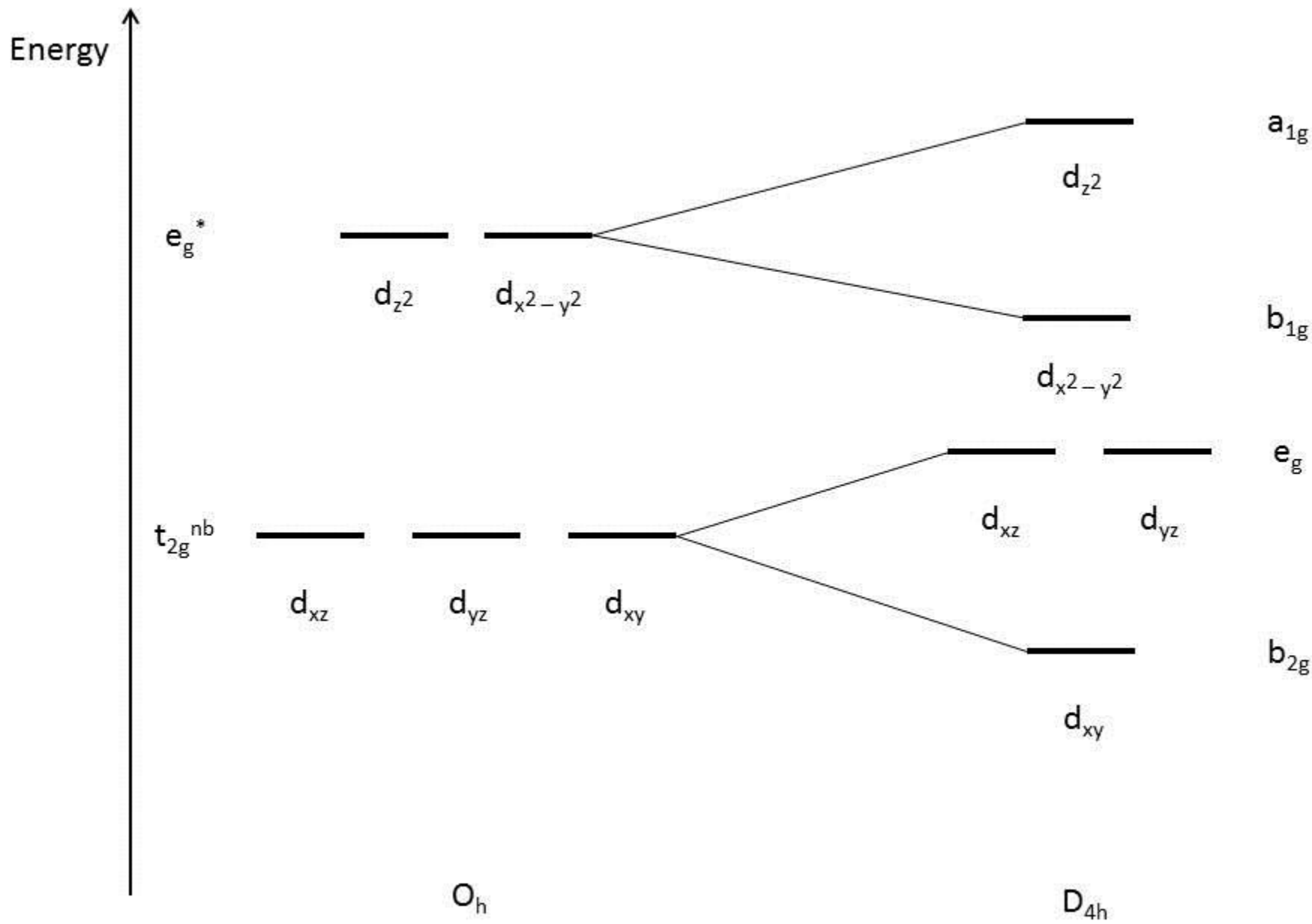


- *Structure of octahedral copper(II) fluoride.*

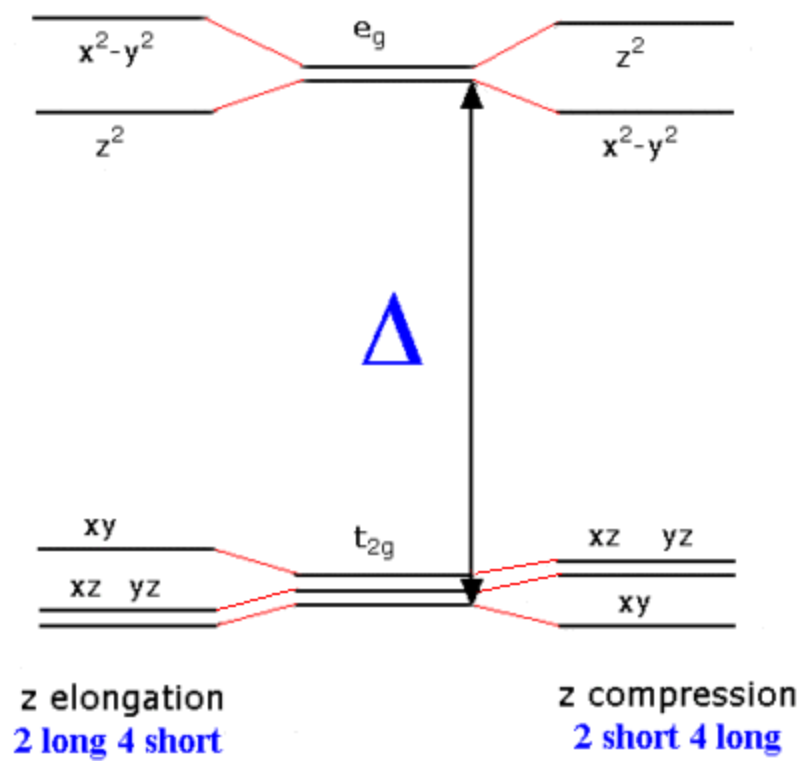
- Notice that the two axial bonds are both elongated and the four shorter equatorial bonds are the same length as each other.
- According the theorem, the orbital degeneracy is eliminated by distortion, making the molecule more stable .

Compression

- Compression Jahn-Teller distortions occur when the degeneracy is broken by the stabilization (lowering in energy) of the d orbitals *without* a z component, while the orbitals with a z component are destabilized (higher in energy) as shown below:



- *→ The Illustration of tetragonal distortion (compression) for an octahedral complex.*
- This is due to the z-component d orbitals having greater overlap with the ligand orbitals, resulting in the orbitals being higher in energy.
- Since the d_z^2 orbital is antibonding, it is expected to increase in energy due to compression.
- The d_{xz} and d_{yz} orbitals are still nonbonding, but are destabilized due to the interactions.



- **The Jahn-Teller Theorem does NOT say how large a distortion should occur.**

Take for example, the 3 copper complexes shown below:

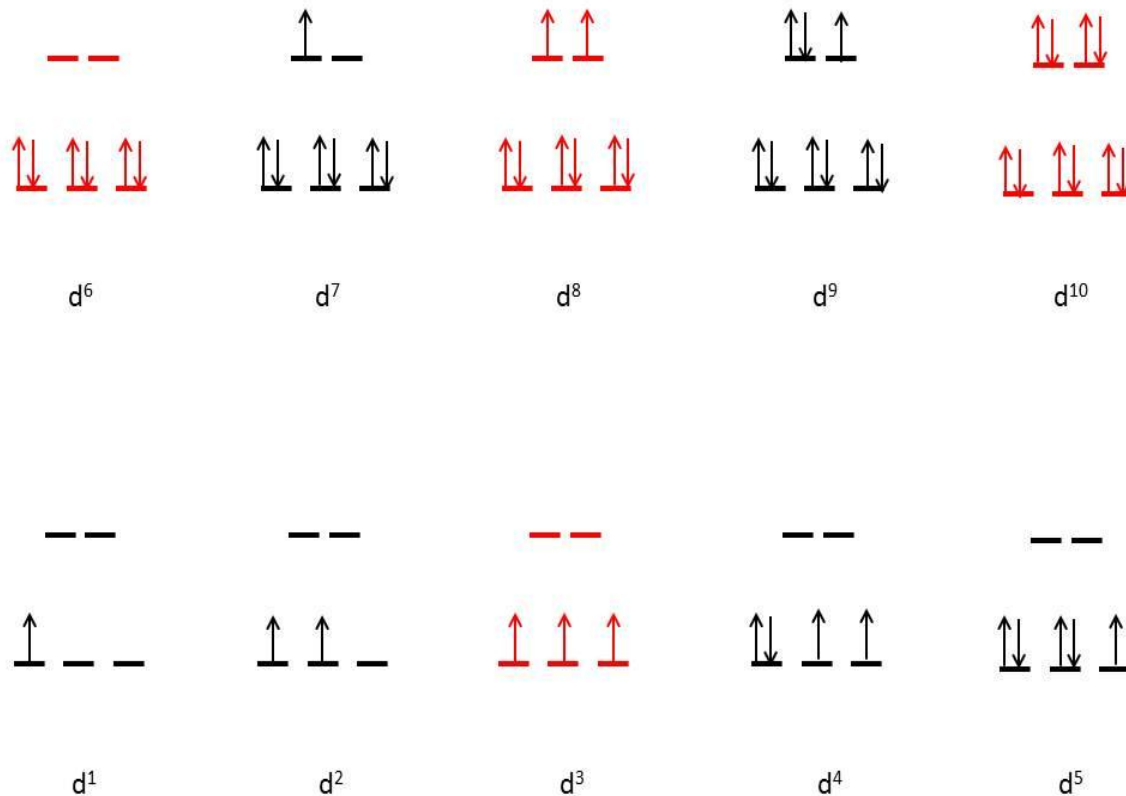
1. tris(octamethylmethylenediphosphonic diamide)copper(II) perchlorate
2. hexakis(pyridine N-oxide)copper(II) dinitrate dihydrate
3. tris(octamethylpyrophosphoramidate)copper(II) perchlorate

- example includes complexes of the type $M_2PbCu(NO_2)_6$.
For M=Cs, below 285 K the molecule shows tetragonal symmetry, for M=K this occurs at below 273 K, for M=Rb at less than 276 K and for M=Tl at temperatures less than 245K.
- Above these temperatures the molecules appear octahedral due to the dynamic Jahn-Teller effect.

Electronic Configurations

- For Jahn-Teller effects to occur in transition metals there must be degeneracy in either the t_{2g} or e_g orbitals.
- The electronic states of octahedral complexes are classified as either low spin or high spin.
- The spin of the system is dictated by the chemical environment.
- This includes the characteristics of the metal center and the types of ligands.

- **Low Spin**
- The following shows the various electronic configurations for low spin octahedral complexes:

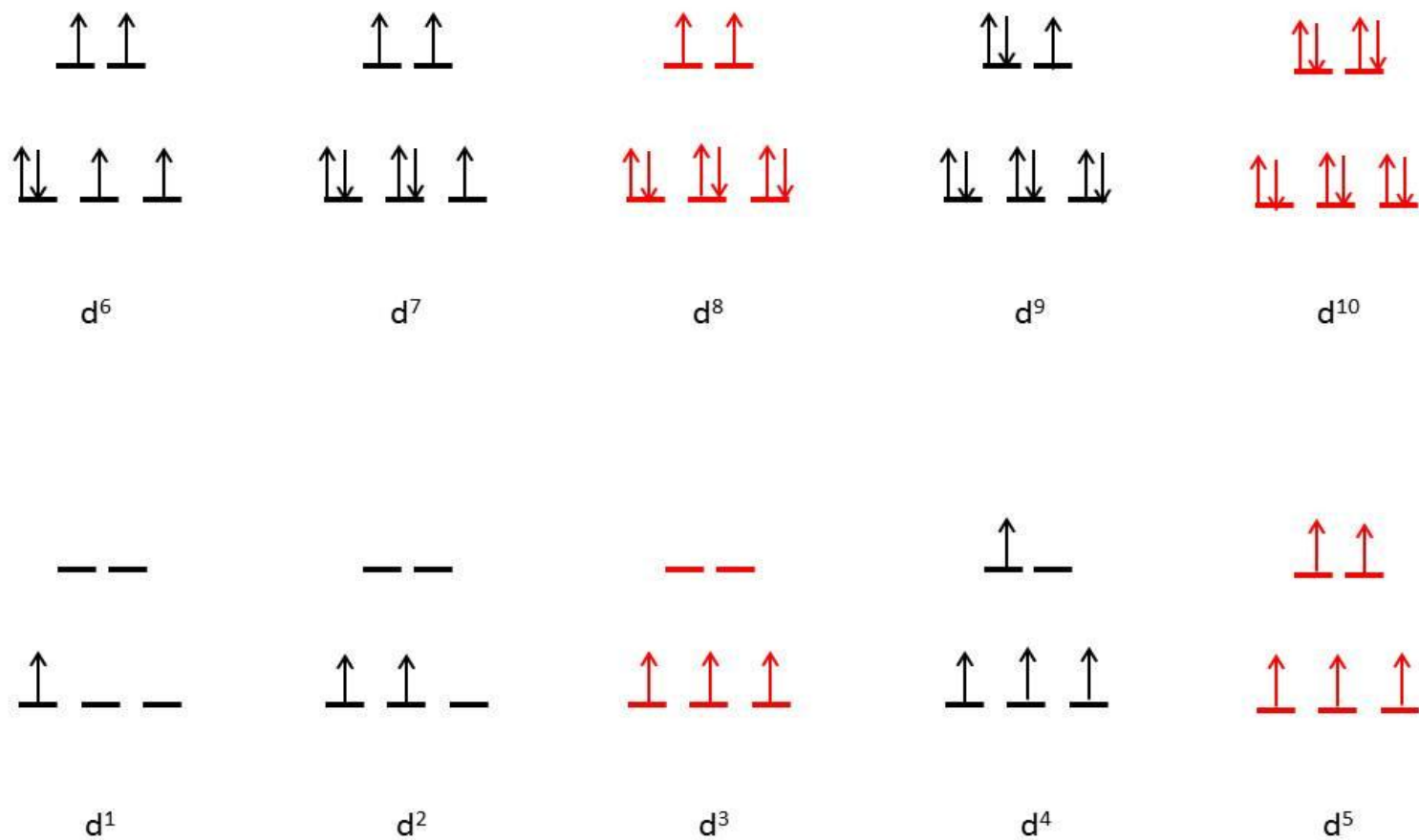


Low spin octahedral coordination diagram (red indicates no degeneracies possible, thus no Jahn-Teller effects).

- The figure illustrates that low spin complexes with d^3 , d^5 , d^8 , and d^{10} electrons that do not exhibit Jahn-Teller distortions.
- These electronic configurations correspond to a variety of transition metals.
- common examples include Cr^{3+} , Co^{3+} , and Ni^{2+} .

High Spin

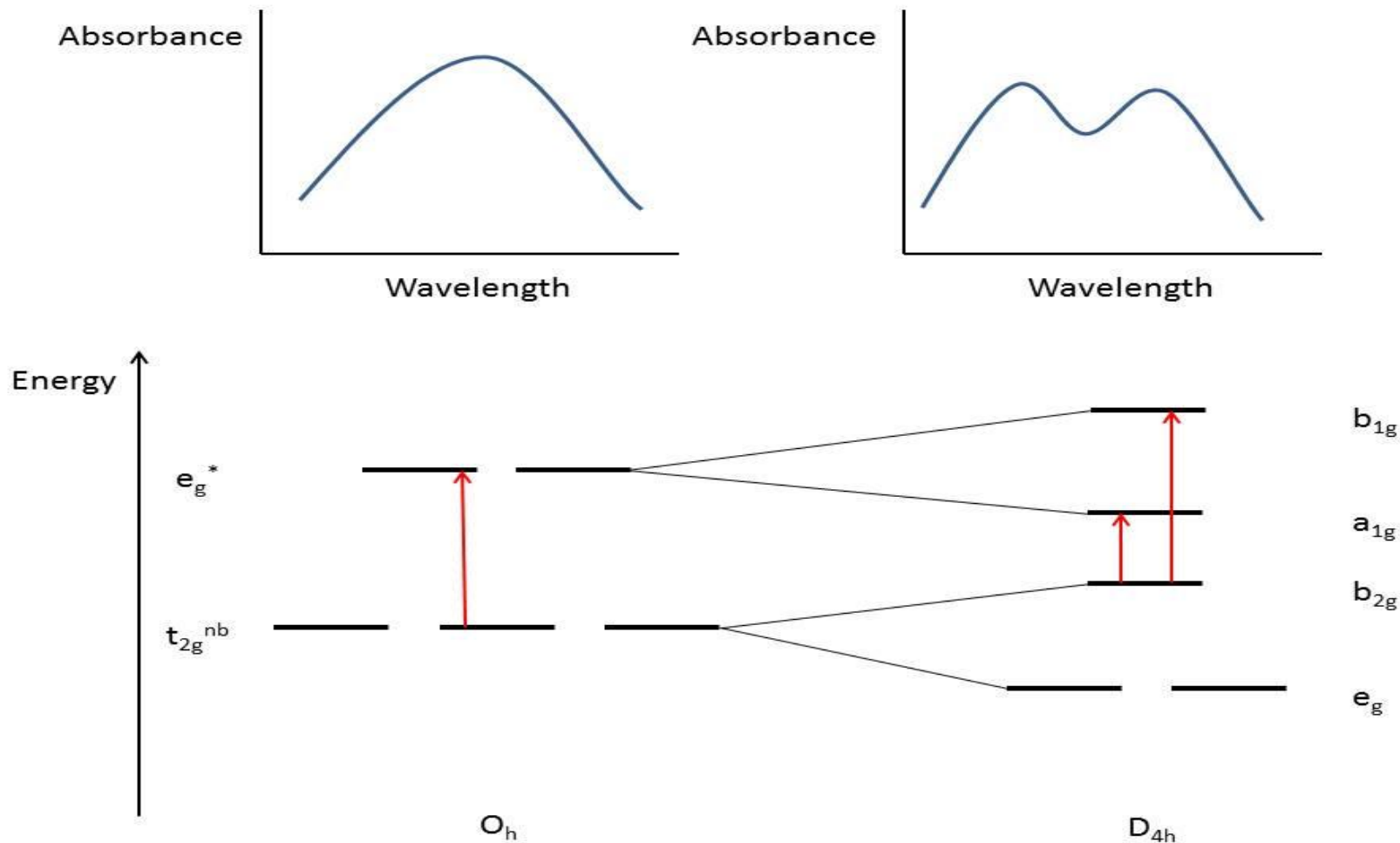
- **Below** shows the various electronic configurations for high spin octahedral complexes:



High spin octahedral coordination diagram (red indicates no degeneracies possible, thus no Jahn-Teller effects).

- The figure illustrates that low spin complexes with d^3 , d^5 , d^8 , and d^{10} electrons cannot have Jahn-Teller distortions.
- In general, degenerate electronic states occupying the eg orbital set tend to show stronger Jahn-Teller effects.
- This is primarily caused by the occupation of these high energy orbitals.
- Since the system is more stable with a lower energy configuration, the degeneracy of the eg set is broken, the symmetry is reduced, and occupations at lower energy orbitals occur

- Jahn-Teller distortions can be observed using a variety of spectroscopic techniques.
- In UV-VIS absorption spectroscopy, distortion causes splitting of bands in the spectrum due to a reduction in symmetry (O_h to D_{4h}).
- Consider a hypothetical molecule with octahedral symmetry showing a single absorption band.
- If the molecule were to undergo Jahn-Teller distortion, the number of bands would increase as shown :



Hypothetical absorption spectra of an octahedral molecule (left) and the same molecule with Jahn-Teller elongation (right). The red arrows indicate electronic transitions.

- A similar phenomenon can be seen with IR and Raman vibrational spectroscopy.
- The number of vibrational modes for a molecule can be calculated using the $3n - 6$ rule (or $3n - 5$ for linear geometry) rule.
- If a molecule exhibits an O_h symmetry point group, it will have fewer bands than that of a Jahn-Teller distorted molecule with D_{4h} symmetry.
- Thus, one could observe Jahn-Teller effects through either IR or Raman techniques.
- This effect can also be observed in EPR experiments as long as there is at least one unpaired electron.

Table 1: Examples of Jahn-Teller distorted complexes

CuBr_2	4 Br at 240 pm 2 Br at 318 pm
CuCl_2	4 Cl at 230 pm 2 Cl at 295 pm
$\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$	2 O at 193 pm 2 Cl at 228 pm 2 Cl at 295 pm
CsCuCl_3	4 Cl at 230 pm 2 Cl at 265 pm
CuF_2	4 F at 193 pm 2 F at 227 pm
$\text{CuSO}_4 \cdot 4\text{NH}_3 \cdot \text{H}_2\text{O}$	4 N at 205 pm 1 O at 259 pm 1 O at 337 pm
K_2CuF_4	4 F at 191 pm 2 F at 237 pm
KCuAlF_6	2 F at 188 pm 4 F at 220 pm
CrF_2	4 F at 200 pm 2 F at 243 pm
KCrF_3	4 F at 214 pm 2 F at 200 pm
MnF_3	2 F at 209 pm 2 F at 191 pm 2 F at 179 pm

- The Jahn-Teller Theorem predicts that:
- distortions should occur for any degenerate state, including degeneracy of the t_{2g} level, however distortions in bond lengths are much more distinctive when the degenerate electrons are in the e_g level.

In the electronic spectrum of an aqueous solution of Ti(III), a d^1 octahedral system, the absorption band is not symmetric but rather shows a distinct broad shoulder.

- This has been interpreted in terms of a lowering of the degeneracy of the t_{2g} level and promotion to the excited state occurring to either of the two orbitals, the d_{z^2} and $d_{x^2-y^2}$, which will no longer be degenerate.
- Thus, two transitions are possible but because the energy difference is small, a shoulder appears rather than 2 distinct peaks.

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